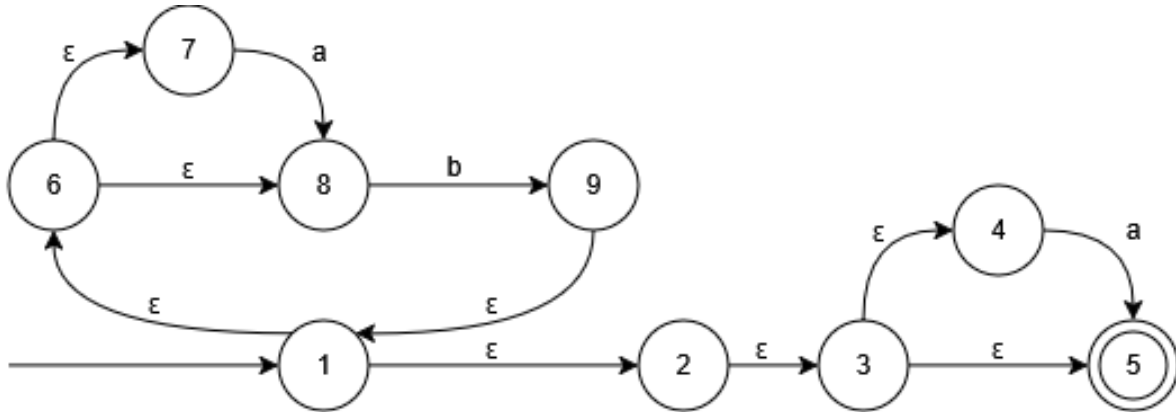


Exercise 3.2

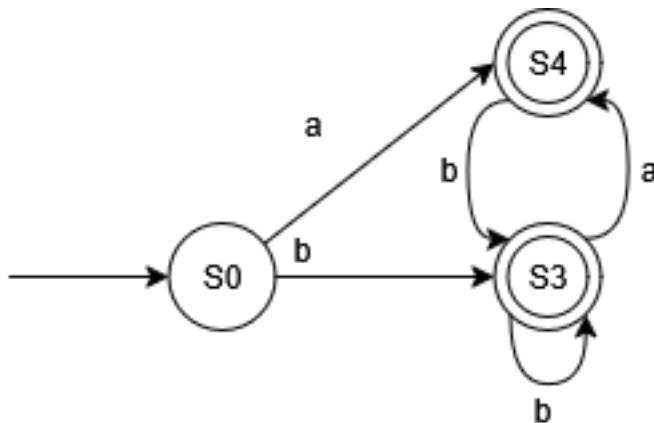
A regular expression that recognizes all sequences consisting of a and b where two a 's are always separated by at least one b can be written as:

$$(a?b)^*a?$$

Below is an NFA of the regular expression:



Below is the NFA is converted into its corresponding DFA:



Exercise 3.3

let $z = (17)$ in $z + 2 * 3$ end EOF

Main

=> Expr EOF	(rule A)
=> LET NAME EQ Expr IN Expr END EOF	(rule F)
=> LET NAME EQ EXPR IN EXPR PLUS EXPR END EOF	(rule H)
=> LET NAME EQ EXPR IN EXPR PLUS EXPR TIMES EXPR END EOF	(rule G)
=> LET NAME EQ EXPR IN EXPR PLUS EXPR TIMES CSTINT END EOF	(rule C)
=> LET NAME EQ EXPR IN EXPR PLUS CSTINT TIMES CSTINT END EOF	(rule C)
=> LET NAME EQ EXPR IN NAME PLUS CSTINT TIMES CSTINT END EOF	(rule B)
=> LET NAME EQ LPAR Expr RPAR IN NAME PLUS CSTINT TIMES CSTINT END EOF	(rule E)
=> LET NAME EQ LPAR CSTINT RPAR IN NAME PLUS CSTINT TIMES CSTINT END EOF	(rule C)

Exercise 3.4

